



QUESTION BANK

Course Code: CY25CO1 Course Name: APPLIED CHEMISTRY – I Class: I (Common to all)

UNIT I – WATER TECHNOLOGY

Water quality parameters and standards. Industrial feed water, Remediation. Municipal water treatment. Desalination.		
Q.No	Questions	Knowledge Level
PART- A (2 marks)		
1.	Write the significance of colour in water.	K2
2.	State term of taste and odour of water.	K2
3.	How can hardness of water be detected?	K2
4.	Mention the significance of feed water.	K2
5.	Write term of hardness of water.	K2
6.	Distinguish between carbonate and non-carbonate hardness.	K2
7.	State term of hard water and soft water.	K2
8.	Mention the units of hardness. How hardness is expressed in terms of CaCO_3 equivalent?	K3
9.	Write down the causes for alkalinity of water.	K2
10.	Any two disinfection methods for municipal water treatment.	K2
11.	State term of break point of chlorination	K2
12.	Write the reasons for caustic embrittlement in boilers.	K2
PART- B (16 marks)		
1.	What are the water quality parameters? Explain their significance.	K2
2.	Write the detailed description about the municipal water treatment process.	K2
3.	Explain the treatments involved in boiler feed water by the important internal conditioning aspects.	K2
4.	With necessary diagram, describe the reverse osmosis method for the desalination of brackish water.	K3
5.	Explain the industrial feed water with suitable examples.	K2
6.	How is water softened by ion exchange method? Explain in detail with a neat diagram.	K2
7.	Describe the principle and procedure involved in the zeolite process for water treatment.	K2

Signature of the staff

Signature of the HoD



QUESTION BANK

Department of Computer Science and Engineering

Course Code: CY25CO1 Course Name: APPLIED CHEMISTRY – I

Class: I CSE

UNIT II – NANOCHEMISTRY

Classification, Size dependent properties. Preparation of nanomaterials, Top-down and Bottom - Up approaches, Applications.		
Q.No	Questions	Knowledge Level
PART- A (2 marks)		
1.	List out the uses of nanoparticles and nanorods.	K2
2.	Mention the applications of nano cluster and nano wires.	K2
3.	Why nanomaterials are size dependent?	K2
4.	Write any properties of nanomaterials with a suitable example.	K2
5.	State term of nanoparticles. Give any two examples.	K2
6.	List out the application of nanoclusters.	K2
7.	Give the important application and properties of nanoclusters	K2
8.	Mention any four applications of nanorods and nano wires.	K2
9.	Differentiate electro deposition and electro spinning.	K2
10.	List out the types of carbon nanotubes.	K2
11.	Write about the important categories of nanomaterials	K2
PART- B (16 marks)		
1.	Explain the size-dependent properties of nanomaterials.	K2
2.	Discuss the types of carbon nanotubes and their applications.	K2
3.	Discuss in a detailed manner about the properties and uses of nanoclusters and nanowires.	K2
4.	With a neat sketch, explain sol-gel synthesis for producing nano materials.	K3
5.	Explain the preparation of nanomaterials by electro deposition and Chemical vapour deposition.	K2
6.	Explain the applications of nanomaterials in medicine, agriculture, energy and catalysis with examples.	K2

Signature of the staff

Signature of the HoD



QUESTION BANK

Department of Computer Science and Engineering

Course Code: CY25CO1 Course Name: APPLIED CHEMISTRY – I

Class: I CSE

UNIT III – ELECTRO – CHEMISTRY

Electrochemical cell, Electrode potential, Redox reaction. Conductivity of electrolytes, Factors.		
Q.No	Questions	Knowledge Level
PART- A (2 marks)		
1.	State term of cell and conductivity.	K2
2.	Give the examples of anode and cathode cells.	K2
3.	Mention the uses of electrochemical cell.	K2
4.	List out the functions of salt bridge in an electrochemical cell.	K2
5.	Write the term of standard electrode potential.	K2
6.	Mention the uses of redox reaction.	K2
7.	How is standard hydrogen electrode (SHE) used as a reference electrode?	K2
8.	Differentiate between oxidation potential and reduction potential with examples.	K2
9.	Differentiate electro deposition and electro spinning.	K2
10.	Give two examples from electrochemical processes.	K2
11.	Write term of conductivity, molar conductivity, and equivalent conductivity.	K2
12.	List the factors affecting conductivity of electrolytes.	K2
PART- B (16 marks)		
1.	With a neat diagram, explain the construction and working of a Daniell cell.	K3
2.	Write the Nernst equation for electrode potential and explain the terms involved.	K2
3.	Discuss the factors that affect electrode potential.	K2
4.	Explain how electron transfer occurs in a galvanic cell through redox reactions.	K2
5.	With a neat diagram, explain the measurement of specific and equivalent conductance.	K3
6.	Explain with neat diagram, measurement of emf of a cell with significances.	K2

Signature of the staff

Signature of the HoD



QUESTION BANK

Department of Computer Science and Engineering

Course Code: CY25CO1 Course Name: APPLIED CHEMISTRY – I

Class: I CSE

UNIT IV – CORROSION AND CONTROL

Chemical and electrochemical corrosions, galvanic series, factors influencing corrosion, Electrochemical protection. Organic and Inorganic coating.		
Q.No	Questions	Knowledge Level
PART- A (2 marks)		
1.	Write the characteristics of corrosions.	K2
2.	How corrosions are classified.	K2
3.	Differentiate chemical and electrochemical corrosions.	K2
4.	Write its significance of galvanic series	K2
5.	Differentiate between Electrochemical series and Galvanic series.	K2
6.	Write the term of galvanic corrosion. Give one example.	K2
7.	Write two factors influencing the corrosion rate.	K2
8.	How does temperature affect corrosion?	K2
9.	Differentiate electro deposition and electro spinning.	K2
10.	Give two examples from electrochemical processes.	K2
11.	Write term of conductivity, molar conductivity, and equivalent conductivity.	K2
12.	List the factors affecting conductivity of electrolytes.	K2
PART- B (16 marks)		
1.	What are the preventive methods of controlling chemical and electrochemical corrosion? Explain.	K2
2.	With neat diagrams, explain different types of electrochemical corrosion (hydrogen evolution, oxygen absorption, differential aeration).	K3
3.	When does the electrochemical corrosion occur? Explain the various types of electrochemical corrosion.	K2
4.	How is corrosion prevented by electrochemical methods.	K2
5.	Describe the role of cathodic and anodic areas in galvanic corrosion.	K2
6.	How corrosion can be prevented by organic and inorganic coating methods.	K2

Signature of the staff

Signature of the HoD



QUESTION BANK

Department of Computer Science and Engineering

Course Code: CY25CO1 Course Name: APPLIED CHEMISTRY – I

Class: I CSE

UNIT V – BATTERIES

Conventional, Contemporary and Emerging battery storage technologies, Primary & Secondary Batteries, Battery Pack, Battery Materials, Performance Parameters, Testing, Safety aspects.		
Q.No	Questions	Knowledge Level
PART- A (2 marks)		
1.	Write the term of conventional battery storage systems. Give examples.	K2
2.	List out the advantages and disadvantages of nickel–cadmium batteries.	K2
3.	Any four applications of conventional batteries in industries and households.	K2
4.	State term of primary and secondary batteries with suitable examples.	K2
5.	Differences between primary and secondary batteries.	K2
6.	List three applications each for primary and secondary batteries.	K2
7.	Mention the function of a Battery Management System (BMS) in a pack.	K2
8.	Write the term of battery pack.	K2
9.	How are individual cells arranged (series/parallel) in a battery pack to achieve required voltage and capacity?	K2
10.	Give the examples of capacity, specific energy, and specific power in batteries.	K2
PART- B (16 marks)		
1.	Compare lead–acid and nickel–metal hydride batteries in terms of energy density and cycle life.	K3
2.	Explain the construction and working of Nickel - Cadmium battery.	K2
3.	With a neat diagram explain the charging and discharging of lithium – ion cell.	K3
4.	Describe the conventional, contemporary and emerging battery storage technology.	K2
5.	Explain the various methods for testing and their importance for a battery.	K2
6.	Explain briefly the common battery materials used in battery pack.	K2

Signature of the staff

Signature of the HoD



Mount Zion College of Engineering and Technology

To make Man Whole!!

(An Autonomous Institution)



Approved by AICTE & Affiliated to Anna University. Accredited by NAAC with A+ grade.